LV2 INSTRUMENT FOR THE ISLA S2400 DSP CARD

KeySampler

Record a sound from the S2400 into a DSP bus and play it chromatically from a MIDI keyboard.

Plugin version	0.17 (LV2 minor version 16)
Built for	S2400 DSP Card firmware 3.1.12
Controller preset	Edirol PCR-M30/M50/M80, GM2 set (memory 0)
Manual date	25 October 2026

1 Overview
2 Installation
3 Quick start
4 Recording
5 Banks and slots
6 Playing, envelope and sample points
7 Filter, vibrato and pan
8 MIDI control and CC Learn
9 Using the Edirol PCR-M30
10 Parameter reference
11 Troubleshooting and limitations
A MIDI implementation
B MIDI implementation chart
C Files and storage on the DSP Card

KeySampler is an independent plugin and is not made or supported by ISLA Instruments or Roland. Page references marked "S2400 manual" refer to the ISLA S2400 User Manual (July 2026); "DSP manual" refers to the S2400 DSP Card User Manual (March 2025); "PCR manual" refers to the PCR-M30/M50/M80 Owner's Manual.

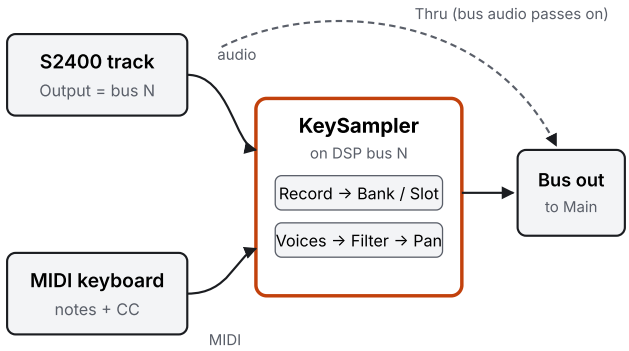
1 Overview

KeySampler sits on a DSP bus like any other plugin. It listens to the audio that the S2400 sends into that bus, records it when you arm it, and then plays the recording across the keyboard: the **Root Note** plays the sound at its original pitch, keys above and below transpose it.

Features

- Records the bus input; starts automatically when sound arrives, stops after 0.4 s of silence (max 30 s)
- 99 banks × 64 slots, stored on the DSP Card and kept through power-off
- 16-voice polyphony, velocity, pitch bend, sustain pedal, portamento and legato glide
- One Shot, Gate and Loop play modes with start and loop points in the S2400's own time format
- ADSR envelope
- Built-in sine, triangle, square and sawtooth oscillators (slots 61–64)
- Resonant 12 dB/oct filter per voice (low, band, high pass) with its own ADSR envelope
- Vibrato with delay, pan, expression
- MIDI CC control with CC Learn, preset for the Edirol PCR-M30 GM2 map
- **Thru** keeps the bus passing its normal audio

Signal flow



The track's audio goes into KeySampler's bus. Thru passes it on unchanged; the filter and pan only affect KeySampler's own notes.

> **Where sounds come from.** The DSP Card cannot read the S2400's SD card or sample memory. It only receives parameter values and, through *Sync Files*, the contents of `SD:/dspcard/`. KeySampler therefore gets its sounds by recording the bus input, or from WAV files placed in its own `samples` folder.

2 Installation

What you need

- An S2400 with the ISLA DSP Card installed
- A computer to copy one folder to the SD card (only for installing or updating the plugin)
- The plugin folder `keysampler.lv2`, containing `keysampler.so`, `keysampler.ttl`, `manifest.ttl` and a `samples` folder

Copy the plugin to the SD card

1. Put the S2400's SD card in your computer.
2. Copy the `keysampler.lv2` folder into `dspcard/lv2/` on the card. The folder must sit directly inside `lv2`, not in an extra sub-folder:

   ```
   SD card / dspcard / lv2 / keysampler.lv2 / keysampler.so
   SD card / dspcard / lv2 / keysampler.lv2 / manifest.ttl  …
   ```

3. When updating, delete the old `keysampler.lv2` folder first so no old files remain.
4. On a Mac, remove the hidden files Finder adds, then eject the card (replace `SDNAME` with the card's name):

   ```
   find /Volumes/SDNAME/dspcard -name ".DS_Store" -delete
   dot_clean /Volumes/SDNAME
   ```

Transfer it to the DSP Card

1. Put the SD card back in the S2400.
2. Press **Shift+Effects** (8th key in the top row), scroll to the **DSP Card** section and select **Sync Files** (DSP manual p. 6). Wait until it finishes.
3. **Switch the S2400 off and on again.** The DSP Card only builds its plugin list at start-up.

Load it on a bus

1. Press **Bank+0** to open the DSP Mixer.
2. Press **Shift+Pad** on a bus to open Bus Settings, select **+Add Plugin/Preset** and choose **KeySampler** (listed under the vendor *KeySampler*).
3. Highlight KeySampler and press **Enter** to open its parameters.

> **32 parameters in the menu.** The S2400 shows only the first 32 parameters of a plugin. KeySampler puts the 32 most-used ones there (chapter 10); the rest keep sensible defaults and several of them can be changed from a MIDI controller.

> **After an update** that adds or changes parameters, remove KeySampler from the bus and add it again, then re-enter its settings in each project that uses it. Recordings are not affected.

3 Quick start

This records a pad and plays it from the keyboard in about a minute.

1. **Choose the bus.** Load KeySampler on bus 1 (see *MIDI routing* below).
2. **Route the pad to it.** Hold **Shift** and press the pad you want to sample to open Track Settings (S2400 manual p. 56). Set **Output** to **1**. Hit the pad: you hear it through KeySampler because **Thru** is on.
3. **Pick a slot.** In KeySampler's parameters set **Bank** to this project's number and **Slot** to an empty slot, e.g. 2.
4. **Record.** Turn **Record** on (**F1** toggles it), hit the pad once, then turn **Record** off.
5. **Play.** Play the keyboard. Middle C (note 60, the default **Root Note**) plays the sound at its original pitch.

Routing a track to a bus

With the DSP Card installed, each of the S2400's eight outputs feeds the DSP bus with the same number (DSP manual p. 3). A track's **Output** in Track Settings therefore selects the bus. Stereo samples play through an output pair: a stereo sound set to output 1 or 2 uses buses 1 and 2. Make that pair stereo in Bus Settings (**Mix**), or set the track's **Output Mix** to *L+R Mixed* to keep it on one bus (S2400 manual p. 56). In Mixer fader mode the output number of each track is shown above its fader (S2400 manual p. 25).

MIDI routing

KeySampler accepts notes on every MIDI channel. Which bus the S2400 sends keyboard notes to is decided by the S2400. In testing, notes reached **bus 1** and not the others. If you want KeySampler on another bus, try setting your keyboard or MIDI track to the channel with the same number as the bus. If that does not work, keep KeySampler on bus 1 and route the pads you want to sample to output 1.

> **Keep the bus to yourself while recording.** Everything that plays into KeySampler's bus is recorded, so route only the pad you are sampling to that bus during a take.

4 Recording

KeySampler records whatever arrives at its bus input into the current **Bank** and **Slot**.

How a recording starts and ends

Stage	What happens
Armed	Turning **Record** on arms the recorder. Nothing is stored until the input is louder than **Rec Threshold dB** (default −50 dB).
Recording	Starts at the first sample above the threshold, including 32 samples (about 0.7 ms) captured just before it, so the attack is not cut.
Automatic stop	After 0.4 s below the threshold, or when 30 s have been recorded.
Manual stop	Turning **Record** off ends a recording immediately.
Clean-up	Trailing silence is trimmed to 50 ms after the last loud sample and the end gets a short fade-out.
Saved	The recording replaces the contents of the slot and is written to the DSP Card as a 16-bit stereo WAV at the card's sample rate (48 kHz). It is ready to play within a few milliseconds.

To record another take, turn **Record** off and on again. Turning it on while a recording is being saved has no effect.

Threshold

- If background noise or a reverb tail keeps the recording running (the slot stays silent until it ends), raise **Rec Threshold dB**, or turn Record off to end the take.
- For very quiet sounds, lower it towards −80 dB.

Thru

Thru (default on; not in the S2400 menu, switched by a learned controller: CC Learn **Thru**) passes the bus input to the bus output unchanged, so the bus behaves normally. With Thru off, the bus only outputs KeySampler's notes. Thru does not affect recording.

> **Mono buses** record mono; a stereo bus pair records stereo. The recorded level is the bus input level before KeySampler's Level, filter and pan.

5 Banks and slots

Think of a cabinet on the DSP Card: **Bank** chooses a drawer (1–99) and **Slot** chooses one of its 64 compartments. Each compartment holds one sound. Bank and Slot together decide which sound plays and where a new recording goes.



Bank 1 / Slot 2 is selected: it plays, and the next recording goes into it.

Situation	Behaviour
Playing	One sound at a time per KeySampler: the current Bank + Slot. Changing Bank or Slot switches immediately and stops notes that are still sounding.
Recording	Always into the current Bank + Slot, replacing what was there. Other slots and banks are never touched.
Slots 61–64	Built-in oscillators in every bank: **61 Sine**, **62 Triangle**, **63 Square**, **64 Sawtooth**. They cannot be recorded over (Record is ignored on these slots).
Empty slot	Plays the matching WAV from the plugin's `samples` folder (slot N = Nth file by name), otherwise silence. Slot 1 holds the test tone `00test.wav` in every bank until you record over it.
Storage	On the DSP Card's internal storage. Recordings survive power-off, Sync Files and plugin updates. They are not on the SD card, so they are not part of an SD-card backup and do not move with a project to another S2400.

Oscillator slots

Slots 61–64 make KeySampler a simple synth. Root Note plays middle C (261.63 Hz); the waveforms are band-limited per octave, so high notes stay clean. They sustain while a key is held and stop on release whatever the Mode; Start and loop points don't apply. Everything else (envelopes, filter, vibrato, glide, CC control) works as usual.

One bank per project

Bank and Slot are parameters, and the S2400 saves parameters with each project, so loading a project switches KeySampler back to that project's bank. Give each project its own Bank number (project 1 = Bank 1, project 2 = Bank 2 ...). New projects start on Bank 1: set a free bank before recording. Two projects on the same bank share their sounds, which is handy for a common kit.

> **Why not automatic?** The S2400 saves only parameter values in a project and never tells the DSP Card the project name, so a parameter (Bank) is the reliable way to keep projects apart.

Adding WAV files with a computer

Copy WAVs into `SD:/dspcard/lv2/keysampler.lv2/samples/` and run **Sync Files**. Files are numbered alphabetically, so start names with numbers (`01PIANO.wav` , `02BASS.wav`). Supported: 8/16/24/32-bit PCM and 32/64-bit float, mono or stereo, any sample rate. Loop points stored in the WAV (smpl chunk) are used in Loop mode. The S2400's file browser does not show the `dspcard` folder, so this needs a computer. Several KeySamplers on different buses have independent Bank and Slot settings.

6 Playing, envelope and sample points

Playing

Parameter	Description
Level	Output volume of KeySampler's notes, 0–2 (1 = unity).
Root Note	The note that plays the sound at its original pitch (default 60 = middle C).
Tune	±24 semitones, fractions allowed (0.5 = 50 cents).
Bend Range	Pitch-bend range in semitones (0–24, default 2). Not in the S2400 menu.
Vel Sens	Not in the S2400 menu (default 1 = full velocity response). 0 = all notes at full volume. Gain = (1 − s) + s × (velocity/127)².
Voices	Polyphony limit, 1–16. Set 1 for mono lines. When all voices are busy, releasing voices are reused first, then the oldest. Replaying a held note retriggers it.
Mode	**One Shot** plays to the end and ignores note-off (drums). **Gate** fades out with Release when the key is let go. **Loop** repeats the loop section while the key is held, then fades out.

The sustain pedal (CC64) holds notes in Gate and Loop modes.

Glide: portamento and legato

Glide Mode	Behaviour
Off	Every note starts at its own pitch.
Portamento	Every new note slides from the previous note's pitch to its own, also when the previous key was already released. Works with any number of Voices; with Voices = 1 the slide starts from wherever the sounding note currently is.
Legato	Mono. While a key is held, a new key does not restart the sound or the envelope: the pitch slides to the new note. Releasing it slides back to the most recent key still held. A note played with no key down starts normally, without a slide.

Glide ms (0–2000, default 100) is the time a slide takes, whatever the interval. In Legato, 0 gives instant pitch changes without retriggering, the classic "hammer-on" feel.

Envelope

Attack ms (0–2000), **Decay ms** (0–5000), **Sustain** (level 0–1) and **Release ms** (0–5000) shape each note's volume. Attack and decay are linear ramps; release fades from the current level. With Sustain 0 in Gate or Loop mode, a note ends after its decay.

Start and loop points

Each point uses two whole-number parameters, **sec** and **ms**, matching the S2400's waveform editor, which shows points as minutes:seconds.milliseconds+samples (S2400 manual p. 30). The sample-accurate "smp" parts exist but sit beyond the S2400's 32-parameter menu, so in the menu points move in 1 ms steps.

S2400 shows	KeySampler parameters
`0:01.250+24`	**sec** = 1, **ms** = 250 (the +24 samples are dropped)
`0:00.500+0`	**sec** = 0, **ms** = 500

- **Start** sets where every note begins.
- **Loop Start** and **Loop End** set the section repeated in Loop mode.

- Loop End at 0:00.000+0 means *end of the sound*; so does any value past the end.
- With Loop Start and Loop End both at 0, a WAV's own loop points are used if it has any, otherwise the whole sound loops.
- Samples are counted at the sound's own sample rate (48 per ms for recordings).

> **Editing tips.** Each encoder click changes a whole-number parameter by one step. For big moves, highlight the parameter and use **Fader 8**, or **F1**/**F2**/**F3** for 25/50/75 % (DSP manual p. 11). A knob assigned to Start, Loop Start or Loop End sweeps across the whole sound, which is quick for finding a loop by ear before fine-tuning it here. If a loop clicks, nudge its points by a few ms or samples.

7 Filter, vibrato and pan

These affect only KeySampler's notes, never the Thru signal.

Filter

Parameter	Description
Filter	Off, Low Pass, Band Pass (mid) or High Pass. A 12 dB/octave state-variable filter on each note.
Cutoff	0–100, spread musically: 0 = 20 Hz, 50 ≈ 630 Hz, 100 = 20 kHz (frequency = 20 × $1000^{Cutoff/100}$ Hz). Changes glide over about 10 ms, so sweeps are smooth.
Resonance	0–1 (Q from 0.7 to about 18). At 0.9 the cutoff frequency is boosted by about 14 dB.

Filter envelope

The filter runs separately on every note, so each note can have its own sweep. A second envelope moves the cutoff: cutoff = **Cutoff** + envelope × **F.Env Amount**. **F.Env Amount 0 switches it off** (the default).

Parameter	Description
F.Env Amount	−100 to +100, in Cutoff units; 0 = off. +60 with Cutoff 20 opens the filter to 80 at the envelope's peak; negative values close it instead.
F.Env Attack / Decay ms	Time to reach the peak (0–2000 ms) and to fall to the sustain level (0–5000 ms).
F.Env Sustain	Level held while the key is down, 0–1.
F.Env Release ms	Time to fall to zero after release, 0–5000 ms.

The filter envelope restarts with each new note, except legato notes, which keep the running sweep. It only acts when **Filter** is not Off. Typical pluck: Low Pass, Cutoff 20, Resonance 0.3, Amount +60, Attack 5, Decay 400, Sustain 0.

Vibrato

Parameter	Description
Vib Rate Hz	Speed, 0.1–12 Hz (default 5). Not in the S2400 menu: set it with PCR knob R3 or CC Learn.
Vib Depth cents	Pitch swing, 0–100 cents either side (default 0 = off).
Vib Delay ms	Time after note-on before vibrato starts, 0–2000 ms. Vibrato then fades in over 200 ms. Not in the S2400 menu: PCR knob R5 or CC Learn.

Pan and expression

Pan (−1 left to +1 right) is a balance control: the centre leaves both sides at full level. **It only works on a stereo bus pair**: in Bus Settings set **Mix** to stereo for the pair KeySampler is on (1&2, 3&4 ...; DSP manual p. 9). On a mono bus there is only one channel to send the sound to. The expression pedal (CC11) scales KeySampler's output from 0 to full; it starts at full.

8 MIDI control and CC Learn

Many parameters can be controlled from MIDI controllers. The assignment table (the *CC map*) is stored on the DSP Card and applies to every project.

Save to Menu and Restore Menu

Moving a knob, slider or button changes **only the sound**. The S2400 menu keeps its values, and those menu values are what the project saves. Two PCR buttons decide what happens to your changes:

Button	What it does
B5 · Save (CC 126)	Writes every value changed from a controller into the S2400 menu. Save the project afterwards to keep them.
B6 · Restore (GM2 System On)	Forgets all controller changes; KeySampler plays the menu values again.

- Changing a value in the menu also takes over that one parameter, so the last thing touched is what you hear.
- Save sends the values to the DSP Card's audio engine (sushi), exactly as the menu does. Start and loop points are sent as sec / ms / smp; a Loop End at maximum is sent as 0:00.000+0 (end of sound).
- The S2400 shows the saved values when you re-open the plugin page. Two KeySamplers with identical settings cannot be told apart, and Save does nothing for them.
- **Save to Menu** and **Restore Menu** are also CC Learn functions, so you can put them on any other button (useful if the S2400 does not pass B6's SysEx).
- The hidden parameter **Auto Menu Sync** (off by default) brings back the old behaviour: every controller move is copied to the menu about 0.1 s after it stops.

CC Learn

1. In KeySampler's parameters, set **CC Learn** (the last item in the menu) to the function you want to control.
2. Move the knob, slider or button. The first CC it sends is assigned to that function, and it controls it straight away. (Save to Menu and Restore Menu act on any message from their button, at most every 0.2 s.)
3. Set **CC Learn** back to **Off**. Repeat for other functions.

Each function has one controller; learning a new one replaces the old assignment. Learnable: CC 0–119 except 6, 11, 38, 64, 65 and 98–101 (fixed jobs), plus channel aftertouch. **Reset Map** restores the default (PCR-M30 GM2) assignments.

How controller values are scaled

Function	Controller 0 → 127
Level	0 → 1 (full = unity gain)
Cutoff / Resonance / Sustain	0 → 100 / 0 → 1 / 0 → 1
Attack	0 → 2000 ms, curved (value²) for finer short times
Decay / Release	0 → 5000 ms, curved
Tune / Pan	−12 → +12 semitones / left → right; centre (64) = 0 / middle
Start / Loop Start / Loop End	Beginning → end of the sound
Record / Thru / Filter Env	Off below 64, on from 64
Slot / Filter Type / Mode	1 → 64 / 4 zones / 3 zones, in menu order
Vib Rate / Depth / Delay	0.1 → 12 Hz (curved) / 0 → 100 cents / 0 → 2000 ms
Glide Mode / Glide Time	3 zones (Off, Portamento, Legato) / 0 → 2000 ms, curved
F.Env Amount / A / D / S / R	−100 → +100 (centre 0) / 0 → 2000 ms / 0 → 5000 ms / 0 → 1 / 0 → 5000 ms

9 Using the Edirol PCR-M30

KeySampler's default map follows the PCR-M30/M50/M80 **GM2 SET** (memory 0, the factory setting; PCR manual p. 98). Knobs keep the functions printed on the GM2 template; the sliders carry the envelope and sample points; the buttons switch features.

Ctrl	GM2 template label	Message sent	KeySampler function
R1	Filter Cutoff	`CC 74`	Cutoff
R2	Filter Resonance	`CC 71`	Resonance
R3	Vibrato Rate	`CC 76`	Vibrato rate
R4	Vibrato Depth	`CC 77`	Vibrato depth
R5	Vibrato Delay	`CC 78`	Vibrato delay
R6	Coarse Tuning	`RPN 0/2`	Tune, coarse: semitones (64 = 0)
R7	Fine Tuning	`RPN 0/1`	Tune, fine: ±100 cents
R8	Pan	`CC 10`	Pan
S1	Portamento Time	`CC 5`	Start point
S2	Aftertouch	`Channel pressure`	Loop Start
S3	Envelope Attack	`CC 73`	Attack
S4	Envelope Decay	`CC 75`	Decay
S5	Envelope Release	`CC 72`	Release
S6	Chorus	`CC 93`	Sustain level
S7	Reverb	`CC 91`	Loop End
S8	Volume	`CC 7`	Level
B1	Program Change Dec	`Program change −`	**Record** on / off
B2	Program Change Inc	`Program change +`	Filter: Off → Low Pass → Band Pass → High Pass → Off
B3	Portamento On/Off	`CC 65 (0/127)`	Mode: One Shot → Gate → Loop → One Shot
B4	Poly Mode On	`CC 127`	Glide on / off (back to the last Glide Mode used, Portamento by default)
B5	Mono Mode On	`CC 126`	**Save**: controller changes → menu
B6	GM2 System On	`SysEx F0 7E 7F 09 03 F7`	**Restore**: play the menu values again
P1	Hold	`CC 64`	Sustain pedal
P2	Expression	`CC 11`	Expression (output volume)
L1–L3	Stop / Start / Continue	`Real-time, port 2`	Not used
Lever	Pitch bend / modulation	`Pitch bend / CC 1`	Pitch bend (Bend Range). Modulation is free for CC Learn.

- **B1 and B2** both send program changes; KeySampler tells them apart by direction (lower number = B1, higher = B2). When the PCR is already at 0 or 127 a repeated number still counts: 0 = B1, anything else = B2.
- B1–B4 start from the current setting, including one chosen in the menu. Like knobs, they change the sound only until you press B5 (Save).
- Slot and Thru have no button: choose Slot in the menu, or CC Learn either one onto a spare controller (e.g. the modulation lever).
- R6 and R7 combine: tune = coarse semitones + fine cents. Changing **Tune** in the menu takes over again.
- Glide time is in the menu (**Glide ms**); to put it on a slider, CC Learn **Glide Time** onto S1 (this replaces Start).

> **Check on your S2400.** Notes are known to reach the DSP bus. Whether the S2400 also forwards aftertouch (S2), program change (B1/B2), RPN (R6/R7) and SysEx (B6) has not been confirmed. If one of these controls does nothing, use CC Learn to move its function to a knob or slider that works, or use the menu.

10 Parameter reference

All parameters in menu order. The S2400 shows the first 32 (ports 5–36). "Port" is the LV2 port index; "Default PCR" is the PCR-M30 control mapped by default. Ports 0–4 are MIDI in, audio out L/R and audio in L/R.

Port	Name	Range	Default	Default PCR	Notes
5	Slot	1 – 64	1		61–64 = oscillators
6	Bank	1 – 99	1		One per project
7	Record	Off / On	Off	B1	Arms recording
8	Rec Threshold dB	−80 – −6	−50		Level that starts a recording
9	Mode	1-Shot / Gate / Loop	Gate	B3	
10	Level	0 – 2	1	S8	
11	Root Note	0 – 127	60		Note at original pitch
12	Tune	−24 – 24	0	R6/R7	Fractions allowed
13	Attack ms	0 – 2000	1	S3	
14	Decay ms	0 – 5000	200	S4	
15	Sustain	0 – 1	1	S6	
16	Release ms	0 – 5000	150	S5	
17	Start sec	0 – 59	0	S1	
18	Start ms	0 – 999	0	S1	
19	Loop Start sec	0 – 59	0	S2	
20	Loop Start ms	0 – 999	0	S2	
21	Loop End sec	0 – 59	0	S7	0 = end of sound
22	Loop End ms	0 – 999	0	S7	
23	Filter	Off / LP / BP / HP	Off	B2	12 dB/oct, per note
24	Cutoff	0 – 100	100	R1	20 Hz – 20 kHz
25	Resonance	0 – 1	0	R2	
26	F.Env Amount	−100 – 100	0		0 = off
27	F.Env Attack ms	0 – 2000	5		
28	F.Env Decay ms	0 – 5000	300		
29	F.Env Sustain	0 – 1	0.2		
30	F.Env Release ms	0 – 5000	300		
31	Glide Mode	Off / Porta / Legato	Off	B4	Legato = mono
32	Glide ms	0 – 2000	100		Slide time
33	Vib Depth cents	0 – 100	0	R4	0 = vibrato off
34	Pan	−1 – 1	0	R8	Balance
35	Voices	1 – 16	16		Polyphony
36	CC Learn	Off … Reset Map	Off		See appendix A.7
Not shown in the S2400 menu (beyond its 32 parameters): defaults, reachable from MIDI controllers					
37	Thru	Off / On	On		Pass bus input to output
38	Vib Rate Hz	0.1 – 12	5	R3	
39	Vel Sens	0 – 1	1		Velocity response
40	Bend Range	0 – 24	2	Lever	Semitones
41	Vib Delay ms	0 – 2000	0	R5	Then 200 ms fade-in
42	Start smp	0 – 95	0		Fine-tune in samples
43	Loop Start smp	0 – 95	0		
44	Loop End smp	0 – 95	0		
45	Filter Env	Off / On	On		Master switch for the filter envelope
46	Auto Menu Sync	Off / On	Off		Copy every controller move into the menu (B5 Save works either way)

11 Troubleshooting and limitations

Problem	What to check
KeySampler is not in the plugin list	Run Sync Files, then power-cycle the S2400. Check the folder is `dspcard/lv2/keysampler.lv2/` with the files directly inside, no extra sub-folder, no zip. Remove `.DS_Store` files. Look under the vendor **KeySampler** or **All vendors**.
The pad is not heard through the bus	The track's **Output** must match KeySampler's bus number; **Thru** must be on.
The keyboard plays nothing	Set **Slot** to 1: the test tone should sound. If it does not, the notes are not reaching this bus: use bus 1, or try the MIDI channel matching the bus number.
Record does nothing	Slots 61–64 are oscillators and cannot be recorded over. Choose a slot from 1 to 60.
Recorded but the slot is silent	The recording may still be running (noise or a long tail keeps it open): turn **Record** off. If it never starts, lower **Rec Threshold dB**. Check you are on the same Bank and Slot you recorded into.
A project plays another project's sound	Both projects use the same Bank. Give each project its own Bank and re-record.
Pan has no effect	The bus is mono. Make its pair stereo in Bus Settings (**Mix**).
Loop does nothing	**Mode** must be Loop. Loop End 0 means the end of the sound; check the points are inside the sound's length.
A knob does nothing	Use CC Learn to assign it. If nothing can be learned, the S2400 is not forwarding that type of message.
The menu does not follow a knob	The S2400 redraws the page only when it is opened: leave and re-open it. With two identical KeySamplers, change one setting on one of them so they can be told apart.
Settings changed after an update	Parameters may have been renamed or reordered. Remove and re-add the plugin and re-enter settings.

Known limitations

- Sounds cannot be stored in the S2400 project file; use one Bank per project.
- Recordings live on the DSP Card only (not on the SD card).
- The S2400 file browser does not show the `dspcard` folder, so adding WAV files to the `samples` folder needs a computer. Recording from the bus does not.
- Only bus 1 received keyboard notes in testing.
- After B5 (Save) the S2400 shows the new values only when the plugin page is re-opened.
- Maximum recording length 30 s; maximum point position 59.999 s.

A MIDI implementation

KeySampler receives MIDI from the S2400 through its LV2 event input. It does not transmit MIDI. All messages are accepted on **every channel** (omni). Running status and system real-time messages are handled by the host and are not used.

A.1 Channel voice messages

Message	Status / data	Behaviour
Note off	`8n kk vv`	Releases the note (Gate, Loop). Ignored in One Shot. Release velocity ignored.
Note on	`9n kk vv`	kk 0–127, vv 1–127. vv = 0 is treated as note off. Pitch = kk − Root Note + Tune (semitones).
Polyphonic key pressure	`An kk vv`	Ignored.
Control change	`Bn cc vv`	See A.2–A.4.
Program change	`Cn pp`	Lower number = Record on/off, higher = Filter step (A.5).
Channel pressure	`Dn vv`	Handled as a learnable controller. Default: Loop Start.
Pitch bend	`En ll mm`	14-bit, centre 8192. Range = Bend Range semitones. Applies to all sounding notes.

A.2 Control changes with fixed functions

CC	Name	Behaviour
`6 / 38`	Data entry MSB / LSB	Data for RPN 0/1 and 0/2 (A.4). Other RPNs and NRPNs are ignored.
`11`	Expression	Scales KeySampler's output 0–1 (initial value 127).
`64`	Hold (sustain pedal)	≥ 64 on. Notes released while held keep sounding until the pedal is released.
`98 / 99`	NRPN LSB / MSB	Ignored (not learnable).
`100 / 101`	RPN LSB / MSB	Select the RPN for data entry.
`120`	All sound off	Stops all voices immediately.
`121, 122`	Reset controllers, local	Ignored.
`123`	All notes off	Releases all notes.
`124, 125`	Omni off / on	Ignored (always omni).
`65`	Portamento on/off	Button: steps Mode (One Shot → Gate → Loop), every message. Not learnable.
`126`	Mono mode on	Button: Save to Menu (controller changes → menu). Polyphony is not changed (use Voices).
`127`	Poly mode on	Button: Glide on/off.

A.3 Assignable control changes (default map)

CC 0–119 (except those in A.2) and channel pressure can be assigned with CC Learn. Defaults:

CC	GM2 name	Default function
5	Portamento time	Start point

7	Volume	Level
10	Pan	Pan
71	Resonance (timbre)	Resonance
72	Release time	Release
73	Attack time	Attack
74	Cutoff (brightness)	Cutoff
75	Decay time	Decay
76 / 77 / 78	Vibrato rate / depth / delay	Vibrato rate / depth / delay
91	Reverb send	Loop End
93	Chorus send	Sustain
Dn vv	Channel pressure	Loop Start

Value scaling is listed in chapter 8. A controller value overrides the menu value until that parameter is changed in the menu.

A.4 Registered parameters

RPN (MSB/LSB)	Name	Behaviour
0 / 1	Fine tuning	14-bit via CC6 + CC38, centre 8192 = 0; range ±100 cents.
0 / 2	Coarse tuning	CC6 value − 64 = semitones (40–88 = −24…+24).
127 / 127	RPN null	Deselects; following data entry is ignored.

Coarse and fine combine into Tune; the result replaces the menu Tune value until Tune is changed in the menu.

A.5 Program change

Program change is used for two buttons: a number **lower** than the previous one toggles **Record** (B1, "program change dec"), a **higher** one steps the **Filter** type (B2, "inc"). A repeated 0 counts as lower, any other repeated number as higher; the first program change received counts as lower if it is 0.

A.6 System exclusive

Message	Behaviour
F0 7E 7F 09 03 F7 (GM2 System On)	Restore Menu (forget controller changes).
All other SysEx	Ignored.

A.7 CC Learn targets

0 Off · 1 Level · 2 Cutoff · 3 Resonance · 4 Attack · 5 Decay · 6 Sustain · 7 Release · 8 Tune · 9 Start · 10 Loop Start · 11 Loop End · 12 Slot · 13 Record · 14 Filter Type · 15 Mode · 16 Pan · 17 Vib Rate · 18 Vib Depth · 19 Vib Delay · 20 Thru · 21 Glide Mode · 22 Glide Time · 23 Filter Env · 24 F.Env Amount · 25 F.Env Attack · 26 F.Env Decay · 27 F.Env Sustain · 28 F.Env Release · 29 Save to Menu · 30 Restore Menu · 31 Reset Map

B MIDI implementation chart

Model: KeySampler 0.17 (LV2 plugin for the S2400 DSP Card) · Date: 25 October 2026 · **O** = yes, X = no
* Default assignment, can be changed with CC Learn. Reception depends on what the S2400 forwards to the DSP bus.

Function	Transmitted	Recognized	Remarks
Basic channel			
Default	X	1–16 (all)	Omni
Changed	X	X	
Mode			
Default	X	Mode 1 (omni on, poly)	
Messages	X	X	CC 126/127 are used as buttons, not as mode changes
Note			
Note number	X	0–127	
Velocity: note on	X	**O** (9n v=1–127)	Vel Sens
Velocity: note off	X	X	
Aftertouch			
Polyphonic (key)	X	X	
Channel	X	**O**	Assignable, default Loop Start
Pitch bend			
Pitch bend	X	**O**	0–24 semitones (Bend Range)
Control change			
5	X	**O**	Start point*
6, 38	X	**O**	Data entry (RPN)
7	X	**O**	Level*
10	X	**O**	Pan*
11	X	**O**	Expression
64	X	**O**	Hold
65	X	**O**	Mode step (button)
71, 72, 73, 74, 75	X	**O**	Resonance, Release, Attack, Cutoff, Decay*
76, 77, 78	X	**O**	Vibrato rate, depth, delay*
91, 93	X	**O**	Loop End, Sustain*
0–119 (others)	X	**O**	Via CC Learn (not 65, 98–101)
98, 99 (NRPN)	X	X	
100, 101 (RPN)	X	**O**	RPN 0/1 fine tune, 0/2 coarse tune
120, 123	X	**O**	All sound off, all notes off
126, 127	X	**O**	Save to Menu, Glide on/off
Program change			
Program change	X	**O** (0–127)	Record on/off (lower) / Filter step (higher)
System			
System exclusive	X	**O**	GM2 System On only (Restore Menu)
System common	X	X	
System real time (incl. active sensing, reset)	X	X	
Reset all controllers, local on/off (121, 122)	X	X	

C Files and storage on the DSP Card

Location on the DSP Card	Contents
`/home/isla/user/plugins/lv2/keysampler.lv2/`	The plugin, copied from `SD:/dspcard/lv2/` by Sync Files
`…/keysampler.lv2/samples/`	Optional WAV files; slot N with no recording plays the Nth file by name
`/home/isla/user/keysampler_recordings/slotNN.wav`	Bank 1 recordings
`/home/isla/user/keysampler_recordings/bankBB/slotNN.wav`	Recordings of bank BB (02–99)
`/home/isla/user/keysampler_recordings/cc_map.txt`	Changes to the default CC map (from CC Learn)

- Recordings are 16-bit stereo PCM WAV at the DSP Card's sample rate (48 kHz), about 190 KB per second.
- The recordings folder is outside the plugin folder, so Sync Files and plugin updates do not touch it.
- `cc_map.txt` lists `<cc> <target>` pairs that differ from the defaults (target 0 = unassigned; CC 128 = channel pressure). Deleting it, or choosing **Reset Map**, restores the defaults.

Technical data

Plugin format	LV2 instrument, aarch64, needs glibc 2.17 or newer and the urid:map feature
Ports	1 MIDI in (atom sequence), 2 audio in, 2 audio out, 42 controls
Polyphony	16 voices (1 in Legato), cubic (Hermite) interpolation; glide linear in semitones
Filter	2-pole state-variable (TPT) per voice, LP / BP / HP, 20 Hz – 20 kHz, ADSR envelope
Oscillators	Sine, triangle, square, sawtooth; 2048-point tables, one band-limited table per octave (harmonics < 20 kHz)
Recording	Up to 30 s, threshold −80 to −6 dB, 0.7 ms pre-roll, 0.4 s silence stop
Storage	99 banks × 64 slots
Save to Menu	gRPC (HTTP/2) to sushi on 127.0.0.1:51051: GetAllProcessors, GetProcessorParameters, GetParameterValue, SetParameterValue (normalized values)
Real-time safety	The audio thread never reads files or allocates memory; a background thread loads, saves and rescans the samples folder every 2 s